ACADEMIC YEAR: 2025-26

Course: Skill base Lab Course: Cloud Computing

Course code: CSL605

Year/Sem: TE/VI

Experiment No.: 02
Aim: To create a EC2 instance and host a static web page by binding Elastic IP with remote SSH Service.
Name: Shravani Sandeep Raut
Roll Number: 40
Date of Performance: 07/01/2026
Date of Submission: 12/01/2026

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission.	10	
Total	**20**	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	2
Understanding	4-5	2-3	2
Journal work and timely submission.	8-10	5-8	4

Checked by

Name of Faculty : Dr. Tatwadarshi Nagarhalli

Signature :

Date :

Experiment No. 2

Aim: To create a EC2 instance and host a static web page by binding Elastic IP with remote SSH Service..

Objective:

To launch an Amazon EC2 instance, configure network security, allocate an AWS Elastic IP, and deploy a static website using a web server so that the website can be accessed publicly through the internet.

Theory:
An instance is a virtual server in the AWS Cloud. With Amazon EC2, you can set up and configure the operating system and applications that run on your instance.
When you launch your instance, you secure it by specifying a key pair (to prove your identity) and a security group (which acts as a virtual firewall to control ingoing and outgoing traffic). When you connect to your instance, you must provide the private key of the key pair that you specified when you launched your instance.

An advantage of using Amazon Elastic Compute Cloud (EC2) is the ability to start, stop, create, and terminate instances at any time. However, this flexibility creates a potential challenge with IP addresses. Restarting a stopped instance (or re-creating an instance after another instance is terminated) results in a new IP address. How do you successfully reference a machine when the IP address is constantly changing?
In response to this problem, Amazon offers the ability to allocate an Elastic IP address. An Elastic IP provides you a single IP address that you can associate with different EC2 instances over time. If your EC2 instance has an Elastic IP and that instance is ever stopped or terminated, you can immediately associate a new EC2 instance with the Elastic IP. Your existing applications will not break because the applications see the IP address they were expecting, even though the back-end EC2 instance has changed.
If you're building your own site manually, you should associate an Elastic IP with your enterprise geodatabase instance if you have the geodatabase on its own dedicated instance. Optionally, you can also associate an Elastic IP with your ArcGIS Server instance if you are not already using an Amazon Load Balancer.
If you use ArcGIS Server Cloud Builder on Amazon Web Services to build your site, no Elastic IPs are created. If you want to use SSH or Remote Desktop Connection to connect to instances within your site, you need to create an Elastic IP and assign one to each instance.

If you ever need to stop an instance, you should reassociate it with its Elastic IP after you start the instance again. You can even associate the Elastic IP with a backup instance while the other instance is down. If you don't have an Elastic IP, users' connections to your machines will permanently break if you ever have to stop the instance.

Steps to create a EC2 instance and host a static web page by binding Elastic IP with remote SSH Service:

Step 1: Launch an EC2 Instance and Configure Security

1. **Sign in** to the AWS Management Console.
2. **Navigate** to the EC2 Dashboard and click **"Launch instance"**.
3. **Name** your instance and select an **Amazon Machine Image (AMI)** (e.g., Amazon Linux 2 or Ubuntu Server, which are free-tier eligible).
4. **Choose** an **Instance type**, such as t2.micro (free-tier eligible).
5. **Create a new Key pair** (e.g., in .pem format) and download the file. This is essential for secure SSH access.
6. **Configure Network settings**:
    0. Ensure **"Auto-assign public IP"** is enabled initially (you will replace this with an Elastic IP later).
    0. Create a new **Security group** that allows inbound traffic for:
        - **SSH** (Port 22) from "My IP" or a specific range for secure remote access.
        - **HTTP** (Port 80) from "Anywhere" (0.0.0.0/0) to make your website publicly accessible.

2. **Review and launch** the instance. Wait for the instance state to change to "running".

Step 2: Allocate and Bind an Elastic IP

1. From the EC2 Dashboard, click **"Elastic IPs"** under **"Network & Security"** in the left-hand menu.
2. Click **"Allocate Elastic IP address"**.
3. Keep the defaults (Amazon's pool of IPv4 addresses) and click **"Allocate"**.
4. Select the newly allocated IP address, click the **"Actions"** dropdown, and choose **"Associate Elastic IP address"**.

5. In the association dialog, select your running **EC2 instance** from the dropdown menu and select its private IP address (usually pre-filled).

6. Click **"Associate"** to bind the static IP to your instance. This IP will now persist even if the instance is stopped and started.

Step 3: Connect via SSH and Host the Website

1. In the EC2 instances list, select your instance and copy its new **Public IPv4 address** (which should be your Elastic IP).

2. Open your terminal or SSH client (like PuTTY for Windows).

3. Navigate to the directory where you downloaded your .pem key file and set the correct permissions for secure SSH access:

bash

chmod 400 your-key-pair.pem

4. Connect to your instance using SSH, replacing your-key-pair.pem and your-elastic-ip with your details:

bash

ssh -i "your-key-pair.pem" ec2-user@your-elastic-ip

(Use "ubuntu@your-elastic-ip" for Ubuntu AMI)

5. **Install a web server** (e.g., Apache or Nginx). For Amazon

Linux: bash

sudo yum update -y

sudo yum install httpd

-y

sudo systemctl start httpd

sudo systemctl enable httpd # *(ensures it starts on boot)*

6. **Deploy your static website files**. The default web root directory for Apache on Amazon Linux is /var/www/html/. You can move your files here (e.g., using git clone if your site is in a repository, or scp to transfer files).

bash

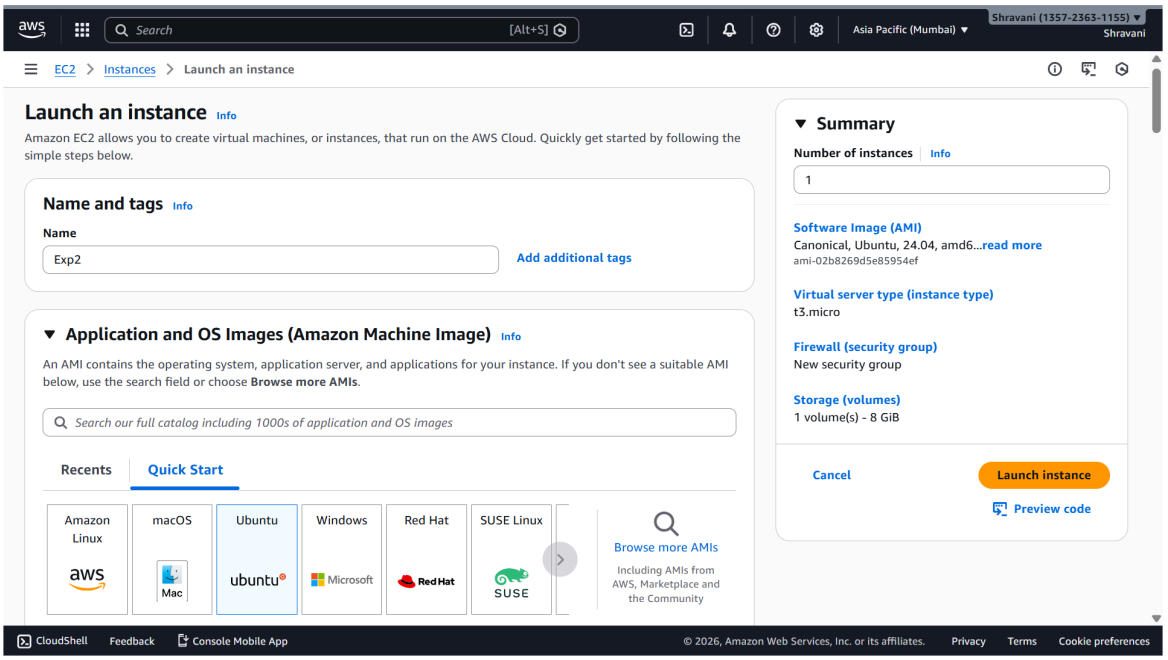
 # Example: create a simple index.html file

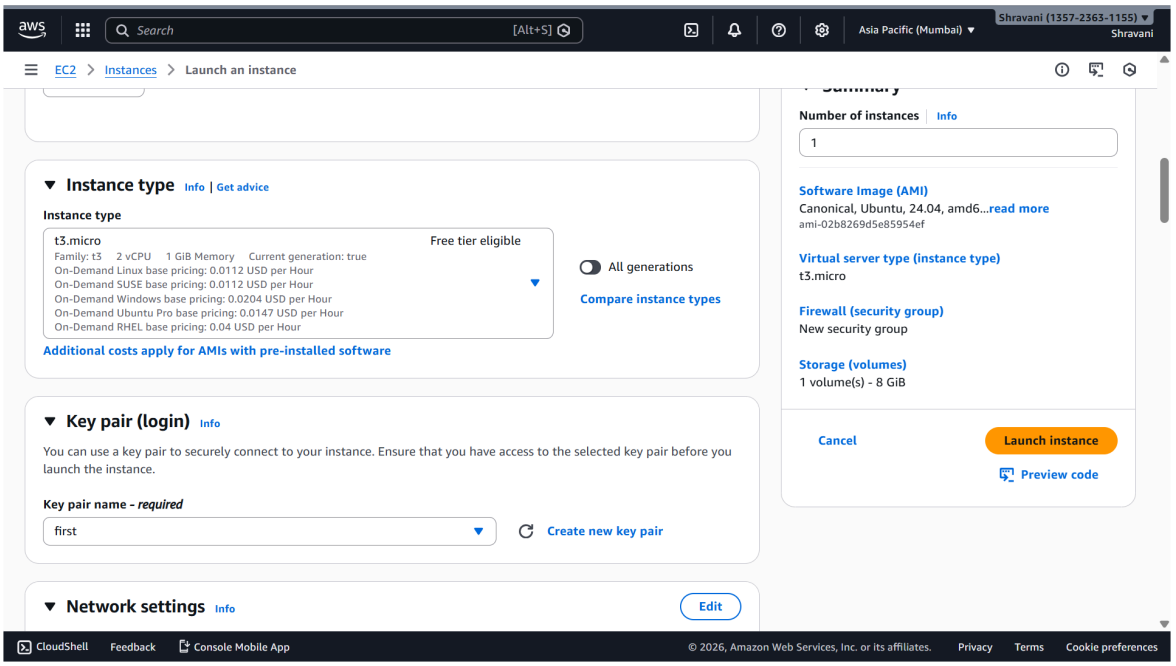
sudo su -

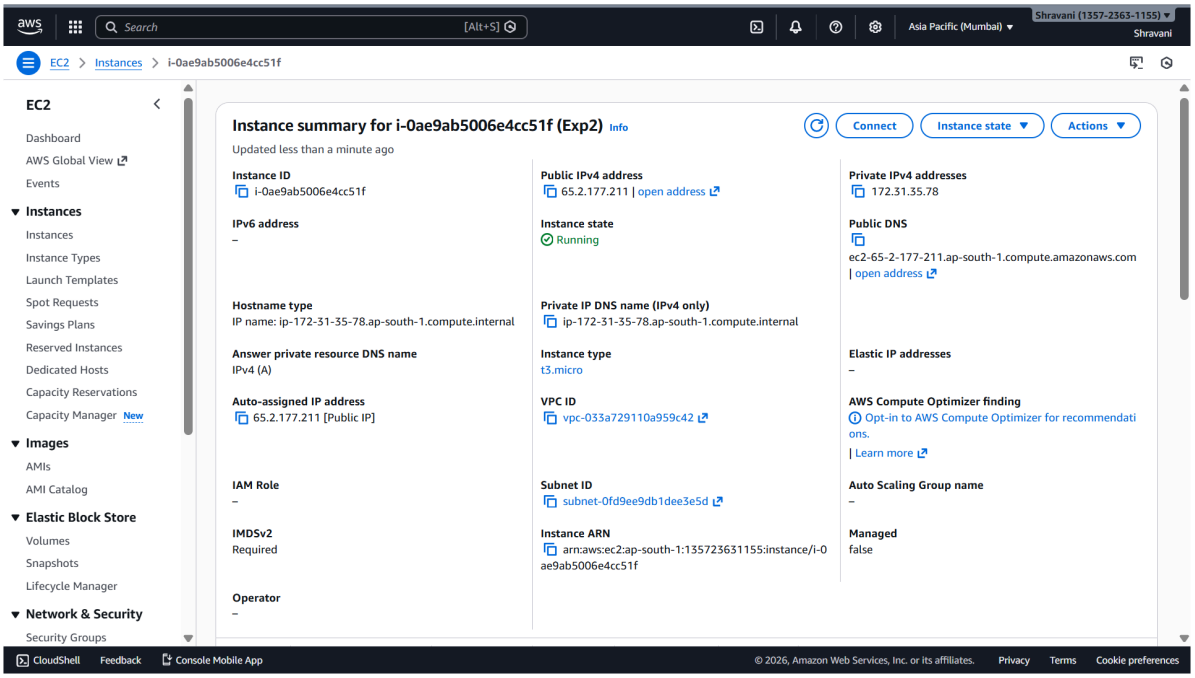
echo "<h1>Hello World! This is my static webpage</h1>" > /var/www/html/index.html

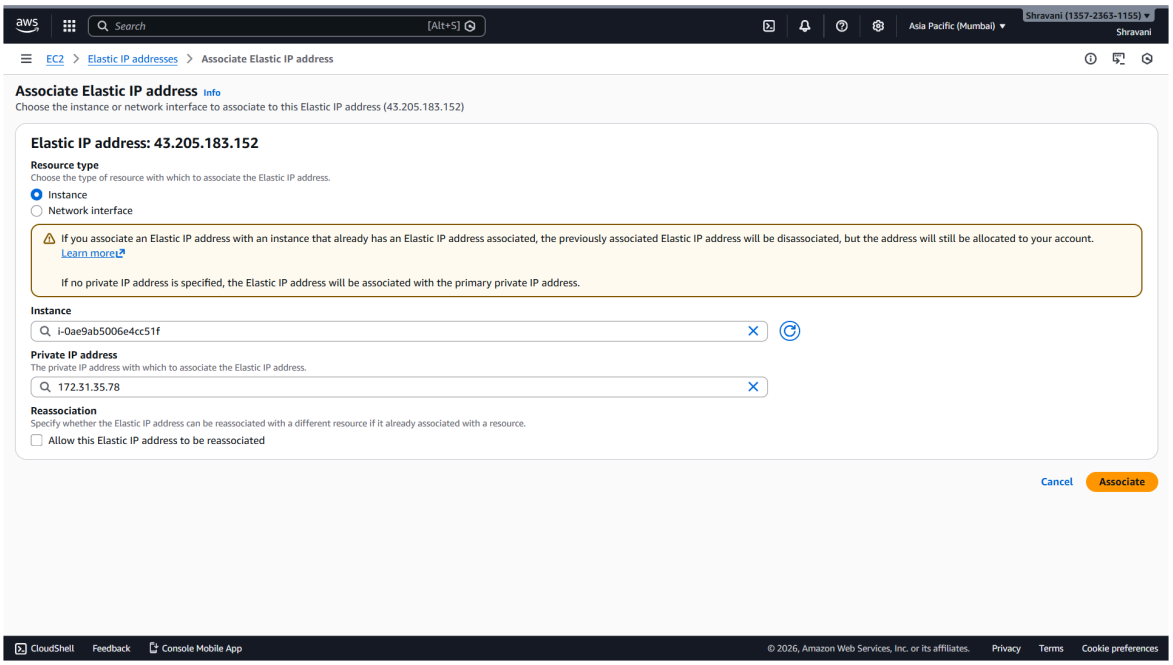
7. **Verify** the website by opening a web browser and entering your Elastic IP address (e.g., http://your-elastic-ip). You should see your static content.

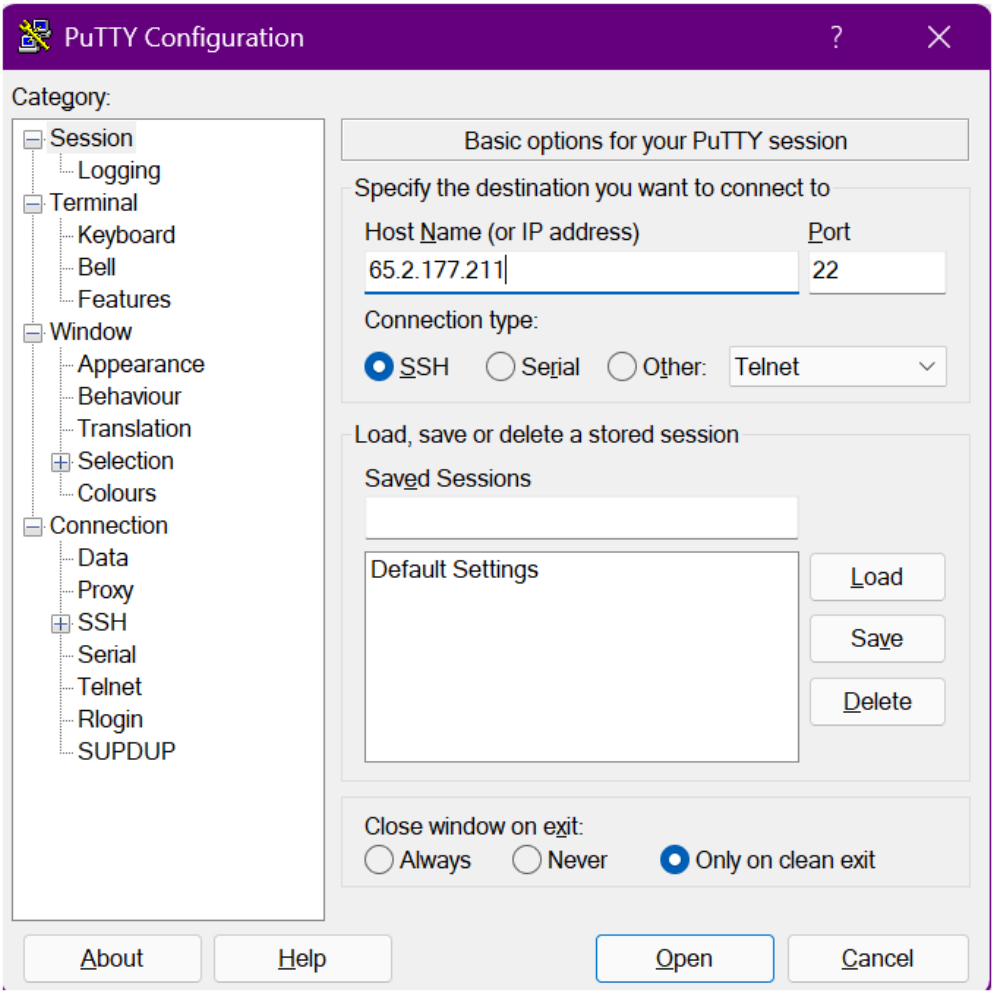
Output /Observation:

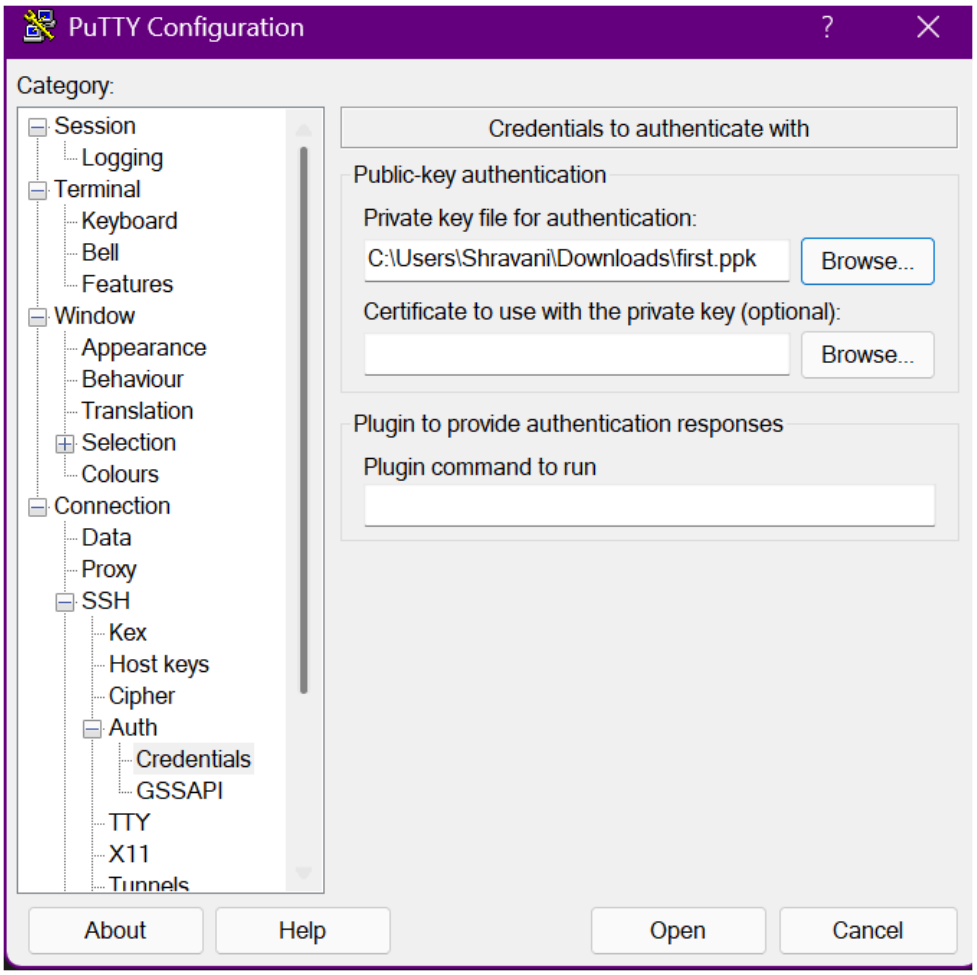





Search [Alt+S]
Asia Pacific (Mumbai)
Shravani (1357-2363-1155)
Shravani
EC2 > Instances > i-0ae9ab5006e4cc51f
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Instance summary for i-0ae9ab5006e4cc51f (Exp2) Info
Updated less than a minute ago
Connect
Instance state
Actions
Instance ID
i-0ae9ab5006e4cc51f
Public IPv4 address
65.2.177.211 | open address
Private IPv4 addresses
172.31.35.78
IPv6 address
–
Instance state
Running
Public DNS
ec2-65-2-177-211.ap-south-1.compute.amazonaws.com | open address
Hostname type
IP name: ip-172-31-35-78.ap-south-1.compute.internal
Private IP DNS name (IPv4 only)
ip-172-31-35-78.ap-south-1.compute.internal
Answer private resource DNS name
IPv4 (A)
Instance type
t3.micro
Elastic IP addresses
–
Auto-assigned IP address
65.2.177.211 [Public IP]
VPC ID
vpc-033a729110a959c42
AWS Compute Optimizer finding
Opt-in to AWS Compute Optimizer for recommendations. | Learn more
IAM Role
–
Subnet ID
subnet-0fd9ee9db1dee3e5d
Auto Scaling Group name
–
IMDSv2
Required
Instance ARN
arn:aws:ec2:ap-south-1:135723631155:instance/i-0ae9ab5006e4cc51f
Managed
false
Operator
–
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


Search [Alt+S]
Asia Pacific (Mumbai)
Shravani (1357-2363-1155)
Shravani
EC2 > Elastic IP addresses > Associate Elastic IP address
Associate Elastic IP address Info
Choose the instance or network interface to associate to this Elastic IP address (43.205.183.152)
Elastic IP address: 43.205.183.152
Resource type
Choose the type of resource with which to associate the Elastic IP address.
Instance
Network interface
If you associate an Elastic IP address with an instance that already has an Elastic IP address associated, the previously associated Elastic IP address will be disassociated, but the address will still be allocated to your account. Learn more
If no private IP address is specified, the Elastic IP address will be associated with the primary private IP address.
Instance
i-0ae9ab5006e4cc51f
Private IP address
The private IP address with which to associate the Elastic IP address.
172.31.35.78
Reassociation
Specify whether the Elastic IP address can be reassociated with a different resource if it already associated with a resource.
Allow this Elastic IP address to be reassociated
Cancel
Associate
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences


PuTTY Configuration
Category:
Session
Logging
Terminal
Keyboard
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Serial
Telnet
Rlogin
SUPDUP
Basic options for your PuTTY session
Specify the destination you want to connect to
Host Name (or IP address)
Port
65.2.177.211
22
Connection type:
SSH
Serial
Other:
Telnet
Load, save or delete a stored session
Saved Sessions
Default Settings
Load
Save
Delete
Close window on exit:
Always
Never
Only on clean exit
About
Help
Open
Cancel


PuTTY Configuration
Category:
Session
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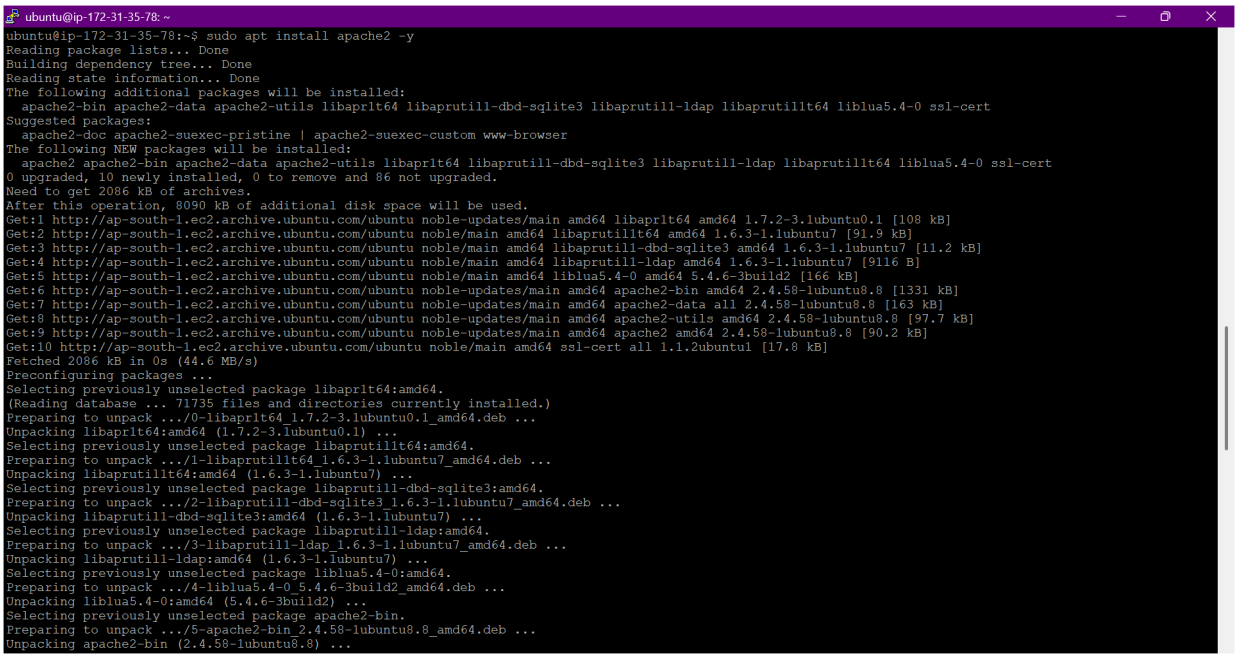
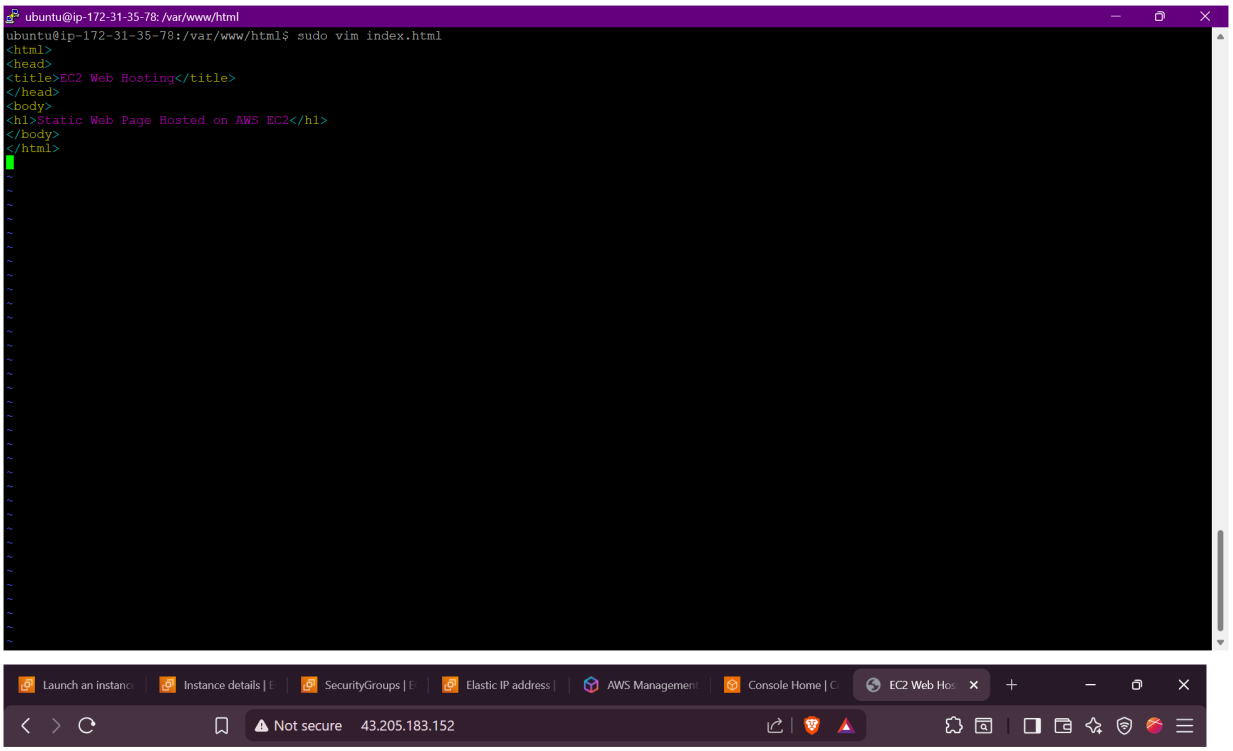
Bell
Features
Window
Appearance
Behaviour
Translation
Selection
Colours
Connection
Data
Proxy
SSH
Kex
Host keys
Cipher
Auth
Credentials
GSSAPI
TTY
X11
Tunnels
Credentials to authenticate with
Public-key authentication
Private key file for authentication:
C:\Users\Shravani\Downloads\first.ppk
Browse...
Certificate to use with the private key (optional):
Browse...
Plugin to provide authentication responses
Plugin command to run
About
Help
Open
Cancel


ubuntu@ip-172-31-35-78: ~
ubuntu@ip-172-31-35-78:~$ sudo apt install apache2 -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
Suggested packages:
apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser
The following NEW packages will be installed:
apache2 apache2-bin apache2-data apache2-utils libapr1t64 libaprutil1-dbd-sqlite3 libaprutil1-ldap libaprutil1t64 liblua5.4-0 ssl-cert
0 upgraded, 10 newly installed, 0 to remove and 86 not upgraded.
Need to get 2086 kB of archives.
After this operation, 8090 kB of additional disk space will be used.
Get:1 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 libapr1t64 amd64 1.7.2-3.1ubuntu0.1 [108 kB]
Get:2 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1t64 amd64 1.6.3-1.1ubuntu7 [91.9 kB]
Get:3 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.3-1.1ubuntu7 [11.2 kB]
Get:4 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 libaprutil1-ldap amd64 1.6.3-1.1ubuntu7 [9116 B]
Get:5 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 liblua5.4-0 amd64 5.4.6-3build2 [166 kB]
Get:6 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-bin amd64 2.4.58-1ubuntu8.8 [1331 kB]
Get:7 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-data all 2.4.58-1ubuntu8.8 [163 kB]
Get:8 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2-utils amd64 2.4.58-1ubuntu8.8 [97.7 kB]
Get:9 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble-updates/main amd64 apache2 amd64 2.4.58-1ubuntu8.8 [90.2 kB]
Get:10 http://ap-south-1.ec2.archive.ubuntu.com/ubuntu noble/main amd64 ssl-cert all 1.1.2ubuntu1 [17.8 kB]
Fetched 2086 kB in 0s (44.6 MB/s)
Preconfiguring packages ...
Selecting previously unselected package libapr1t64:amd64.
(Reading database ... 71735 files and directories currently installed.)
Preparing to unpack .../0-libapr1t64_1.7.2-3.1ubuntu0.1_amd64.deb ...
Unpacking libapr1t64:amd64 (1.7.2-3.1ubuntu0.1) ...
Selecting previously unselected package libaprutil1t64:amd64.
Preparing to unpack .../1-libaprutil1t64_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1t64:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-dbd-sqlite3:amd64.
Preparing to unpack .../2-libaprutil1-dbd-sqlite3_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-dbd-sqlite3:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package libaprutil1-ldap:amd64.
Preparing to unpack .../3-libaprutil1-ldap_1.6.3-1.1ubuntu7_amd64.deb ...
Unpacking libaprutil1-ldap:amd64 (1.6.3-1.1ubuntu7) ...
Selecting previously unselected package liblua5.4-0:amd64.
Preparing to unpack .../4-liblua5.4-0_5.4.6-3build2_amd64.deb ...
Unpacking liblua5.4-0:amd64 (5.4.6-3build2) ...
Selecting previously unselected package apache2-bin.
Preparing to unpack .../5-apache2-bin_2.4.58-1ubuntu8.8_amd64.deb ...
Unpacking apache2-bin (2.4.58-1ubuntu8.8) ...



Static Web Page Hosted on AWS EC2

Conclusion:

1. Why is an AWS Elastic IP used with an Amazon EC2 instance while hosting a website?
   An Elastic IP provides a permanent, static address for your instance. Without it, AWS assigns a new public IP every time the server restarts, which would break your domain's DNS link and take your website offline.

2. How does a web server like Apache HTTP Server help in deploying and accessing a static website on a cloud server?
   Apache acts as the communications bridge. It listens for incoming HTTP requests, retrieves your static HTML files from the server's storage, and delivers them to the user's browser, ensuring your content is accessible over the internet.